

PN1 Practice Test

Neurons and Population Representation

Time: 30 minutes. Total: 30 marks. Work by hand on separate paper and show your reasoning. A basic calculator may be used. This is a practice test, not a course-issued test.

Conventions. The scalar current is $J = \alpha ex + J^{\text{bias}}$, where $e \in \{-1, +1\}$. As in PN1, the named intercept ξ is the threshold in projected input ex : firing begins when $ex = \xi$. Maximum rate occurs at $ex = 1$. In activity matrices, rows are neurons and columns are input samples.

1. Scalar ReLU encoding

[5 marks]

A neuron has $G(J) = \max(0, J)$, $a^{\max} = 120 \text{ Hz}$, $\xi = -\frac{1}{2}$, and $e = -1$.

- (a) Find α and J^{bias} , showing the two current conditions you used. [2]
- (b) At what x -coordinate does it start firing? Explain the sign. [1]
- (c) Give its rates at $x = -1, 0, \frac{1}{2}, 1$, and sketch the tuning curve on $[-1, 1]$. [2]

2. Identity decoding and noise

[7 marks]

Three scalar targets and a two-neuron activity matrix are

$$\mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad A_{\text{noisy}} = \begin{bmatrix} 0 & 1 & \frac{3}{2} \\ \frac{3}{2} & 1 & 0 \end{bmatrix}.$$

Use the clean A to fit decoders. For the noise-aware fit, the specified regularization amount is $m\sigma^2 = 1$, where $m = 3$. Let $\text{RMSE} = \sqrt{\frac{1}{m} \sum_k (x_k - \hat{x}_k)^2}$.

- (a) Compute AA^T and $A\mathbf{x}$, then solve $(AA^T)\mathbf{d} = A\mathbf{x}$. [2]
- (b) Solve $(AA^T + I)\mathbf{d}_{\text{reg}} = A\mathbf{x}$. [1]
- (c) Fill in all four RMSEs for $\hat{\mathbf{x}}^T = \mathbf{d}^T A_{\text{test}}$. You may leave square roots unevaluated. Briefly state the tradeoff. [4]

Decoder	Clean A	Noisy A_{noisy}
Unregularized		
Noise aware		

3. Sources of error

[4 marks]

Illustrative mean-squared errors for different population sizes are:

	n	8	16	32
Distortion error		0.040	0.010	0.0025
Independent-noise error		0.020	0.010	0.0050

- (a) Which row scales roughly as $1/n$, and which as $1/n^2$? Explain using the effect of doubling n . [2]
- (b) If noise standard deviation is divided by 10 with decoders fixed, what happens to noise mean-squared error? Why can independent noise error fall as more neurons are added? [2]

4. LIF rates and decoding**[5 marks]**

For $J > 1$, $G(J) = [\tau_{\text{ref}} - \tau_{\text{RC}} \ln(1 - 1/J)]^{-1}$; otherwise $G(J) = 0$. Use $\tau_{\text{ref}} = 0.002$ s, $\tau_{\text{RC}} = 0.020$ s, $a^{\text{max}} = 100$ Hz, $\xi = 0$, $e = +1$, and $e^{-0.4} \approx 0.6703$.

- (a) Rearrange the rate equation to express J_{max} in terms of a^{max} , then find $J_{\text{max}}, \alpha, J^{\text{bias}}$. Use threshold current $J_{\text{th}} = 1$. [3]
- (b) For what x is this neuron silent? How does the active part of its tuning curve differ from a ReLU line? [1]
- (c) For any LIF activity matrix B with m sample columns and targets $\mathbf{x}$, write the noise-aware decoder normal equation. [1]

5. Two-dimensional tuning**[4 marks]**

A LIF neuron represents $\mathbf{x} = (x_1, x_2)^T$ with $J = 1 + \alpha \langle \mathbf{e}, \mathbf{x} \rangle$, $\alpha > 0$, and $\mathbf{e} = (1/\sqrt{2}, -1/\sqrt{2})^T$. It is silent for $J \leq 1$.

- (a) On the unit disk, give the threshold line and the silent half-disk. State the preferred direction. [2]
- (b) For $\mathbf{x}(\theta) = (\cos \theta, \sin \theta)^T$, write $\langle \mathbf{e}, \mathbf{x}(\theta) \rangle$ as one cosine. Why is the firing-rate curve only approximately cosine-shaped? [2]

6. Vector populations**[5 marks]**

One hundred neurons represent 2D inputs. Let A contain their rates at $m = 1600$ input samples and X contain the input vectors in columns.

- (a) Give the shapes of A, X, D , where $DA \approx X$. Explain the shape of D using a *single* sample of 100 activities. [2]
- (b) Write the noise-aware matrix normal equation for D with independent activity noise of standard deviation σ . [1]
- (c) For a small illustrative population, let encoder columns be $(1, 0)^T, (0, 1)^T, (-1, 0)^T, (0, -1)^T$ and let

$$D = \begin{bmatrix} 1 & -\frac{1}{4} & -1 & \frac{1}{4} \\ -\frac{1}{4} & 1 & \frac{1}{4} & -1 \end{bmatrix}, \quad \mathbf{a} = \begin{bmatrix} 1 \\ \frac{1}{4} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{x}_{\text{target}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute the vector decoded with D and the vector decoded using the encoders as decoders. Which better preserves direction after normalization, and which better preserves magnitude in this example? Why can a broad, uniform set of encoders give a reasonable direction estimate? [2]